

Project Design Phase Proposed Solution Template

Date	10 July 2026
Team ID	SWTID-2026-6119
Project Name	Credit Card Approval Prediction System
Maximum Marks	2 Marks

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Credit card approvals are still worked by hand at most branches, so turnaround time swings from officer to officer, risk gets judged inconsistently, and applicants are left waiting without knowing where they stand.
2.	Idea / Solution Description	A machine-learning-powered web app that reads an applicant's demographic and financial inputs and returns an instant Approved / Rejected call, backed by a model trained on past approval outcomes and served through a lightweight Flask front end.
3.	Novelty / Uniqueness	Rather than committing to one algorithm or a hard-coded rulebook, the project trains and benchmarks several models - Logistic Regression, Decision Tree, Random Forest and XGBoost - and ships whichever one scores best, so the final decision logic is chosen on evidence rather than assumption.
4.	Social Impact / Customer Satisfaction	Applicants get an answer in seconds instead of days of silence, and bank officers gain a dependable second opinion that cuts down repetitive manual review and closes the gap between how different officers judge the same case - a fairer outcome for everyone applying.
5.	Business Model (Revenue Model)	Banks and lenders save on the staff-hours spent reviewing applications by hand and cut losses tied to inconsistent manual calls. The tool can live as an in-house decision aid or be packaged as a subscription credit-scoring module sold to smaller lenders.
6.	Scalability of the Solution	Because preprocessing and modeling are kept as separate, swappable stages, the pipeline can be retrained whenever a larger or fresher dataset arrives. Packaging the Flask app in a container lets it be deployed on cloud infrastructure (e.g., IBM Cloud) and scaled out as application volume grows.